

CoAffinity: A Multimodal Dataset for Cognitive Load and Affect Assessment in Remote Collaboration

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Abstract—Understanding the relationship between cognitive load and affective state in remote work is vital for designing intuitive collaboration. We present CoAffinity, a multimodal dataset encompassing eight structured remote-work tasks, during which 39 participants provided self-reported measures (arousal, valence, positive/negative affect, and cognitive-load) while being recorded via audio, video, and physiological signals (PPG and GSR). Spanning over 38 hours of annotated data, our approach involved precise timestamp alignment, short and long-session labelling, and subsequent machine-learning and deep-learning benchmarks. Key findings show that integrating multiple modalities, especially physiological data, significantly improves the detection of cognitive load and emotion, while group synchrony metrics highlight how physiological coherence shifts under varied task demands. By capturing complex cognitive-emotional dynamics in realistic remote settings, CoAffinity aims to advance affective computing, inform human-computer interaction research, and foster more empathetic remote collaboration tools.

Index Terms—Cognitive load recognition, emotion recognition, empathy, physiological signals, group synchrony, remote collaboration.

I. INTRODUCTION

IN THE rapidly evolving landscape of remote work, understanding the interplay between cognitive load and affective states is important for designing collaborative solutions [1]. Numerous studies have shown that remote work using video conferencing increases mental workload and fatigue [2], [3]. This is due to various factors, including limited physical mobility, challenges in perceiving and transmitting non-verbal cues, prolonged focus on the video conferencing screen, and a lack of social connection. To better understand the nuances of remote work and its impact on individuals, it is important to design and develop tools that can detect and manage cognitive load and

affect in contextual settings. A number of affective computing datasets are available, capturing arousal, valence, and other emotion-related factors [4], [5], [6], [7], [8], [9]. However, these data are not suitable for predicting cognitive load and emotions in remote work settings because the stimuli and context differ from those of video conferencing [10].

The field of affective computing [11] uses wearable technology and artificial intelligence (AI) to track factors associated with affective states, such as emotions and cognitive load [12]. There have been significant advances in hardware, software, and machine learning [13], [14], but the lack of labelled data remains a major bottleneck [13], [15]. This is particularly true for multimodal emotion recognition, which combines various modalities to assess emotion and cognitive load [16], [17]. Currently, there are little to no datasets specifically aimed at understanding users in remote work settings, nor are there audio-visual-physiological datasets targeting cognitive load alongside affective states. Unfortunately, collecting data to train machine learning models is a labour-intensive process with numerous challenges.

To address the growing need for affect-aware remote collaboration, we introduce CoAffinity (**Collaborative-Affinity**), a multimodal dataset designed to capture the complex interplay between cognitive load, affective states, and group synchronization. CoAffinity integrates self-reported measures, audio and video recordings, and physiological signals collected via photoplethysmography (PPG) and galvanic skin response (GSR) sensors, providing a rich foundation for machine learning-based assessment of cognitive and emotional dynamics in remote teamwork. Evaluated through simulated remote work scenarios on a videoconferencing platform, the dataset captures the nuances of collaboration across tasks with varying cognitive and emotional demands [18]. By leveraging audio-video-physiological data from wearable devices, CoAffinity enables the real-time prediction of affective states and cognitive load in contextual remote collaborative settings.

A key dimension of successful collaboration is group synchrony, the alignment of physiological and behavioural responses among team members, which has been linked to enhanced teamwork, shared understanding, and improved performance [19]. Despite its significance, group synchrony remains underexplored in remote settings, where nonverbal cues are often diminished, making it harder to establish interpersonal rapport and cognitive alignment [20]. The CoAffinity study advances this investigation by analysing how synchrony in PPG and

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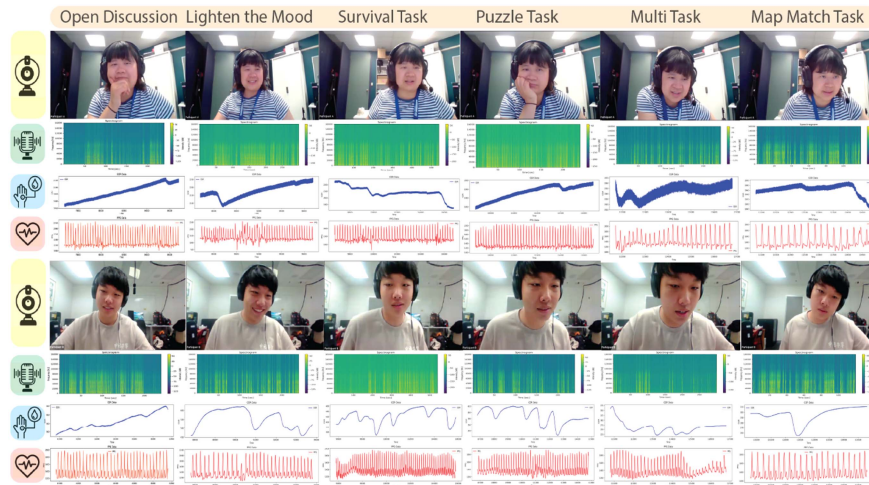


Fig. 1. Sample representative frames across video, audio, PPG, and GSR modalities from two participants completing various tasks in the CoAffinity study.

GSR signals correlates with affective states and cognitive load fluctuations across various collaborative tasks.

The dataset features 39 participants from 8 countries, aged 18 to 40, with a balanced gender ratio. It comprises over 39 hours of recorded multimodal content, yielding more than 1,248 samples, each annotated with task-based self-reported ground truth labels for arousal, valence, positive and negative affect, and cognitive load attributes (mental demand, physical demand, temporal demand, and performance). Fig. 1 visually summarizes the experimental setup, presenting sample frames from different tasks across various modalities alongside self-reported labels, providing an intuitive overview of the study's methodology and participant interactions.

In summary, we make the following contributions:

- 1) We created CoAffinity, one of the first audio-visual and physiological signals datasets that include cognitive load and affect attributes from a remote collaboration setting.
- 2) CoAffinity is an originally recorded audio-visual and physiological signals dataset in English, over 38 hours of videos and physiological signals with self-reported ground truths for affect and cognitive load.
- 3) We investigated group synchronization metrics to assess the significance of physiological signal synchrony across various tasks. Our findings reveal that task-dependent variations in synchrony, with cognitively intense tasks (e.g., Multitask (Active) and Map Matching) exhibiting greater PPG and GSR coherence, whereas conversational tasks (e.g., Open Discussion) demonstrate reduced synchrony, potentially due to asynchronous cognitive processing and independent affective responses.
- 4) We performed an extensive analysis to validate our study design, providing deep learning and machine learning baselines for estimating cognitive load and affect on CoAffinity in both uni-modal and multi-modal setups. Multimodal setups achieved 82.6% F1-score for cognitive load, 80.2% for valence, and 72.1% for arousal.

Following sections (i) reviews publicly available datasets and highlight the novel contributions of CoAffinity (Section II), (ii) details the data collection methodology, including participant recruitment, experimental design, and feature extraction

(Section III), (iii) analyses self-reported labels, examines group synchronization, and validates the dataset using machine and deep learning models, (Sections IV–VI), and (iv) discusses dataset limitations, intended applications, and future directions, concluding with a summary (Section VII).

II. RELATED WORK

Understanding cognitive load and affective states in remote work requires examining existing datasets and their limitations. We analyze these datasets across three key dimensions: emotion recognition, cognitive load assessment, and remote collaboration capabilities.

Emotion Recognition Datasets: Early contributions like DEAP, AMIGOS, Ascertain and DECAF established foundational approaches for emotion analysis using physiological signals and visual content [8], [21], [22]. While these datasets provided valuable insights into affect recognition, they relied primarily on passive content viewing rather than interactive scenarios.

More recent datasets have moved toward naturalistic environments. K-EmoCon and AFF-WILD focus on spontaneous emotional expressions [23], [24], capturing natural social interactions through physiological signals and audiovisual recordings. However, these datasets focus solely on affective states without considering cognitive load measures. The ASCERTAIN dataset [25] connects personality traits with affective states using physiological signals, but it primarily focuses on passive media consumption rather than interactive collaboration. Similarly, the DAPPER dataset [9] provides a large-scale ambulatory assessment of emotion with wearable sensors, yet it lacks structured collaborative tasks or cognitive load measurement, which are central to CoAffinity's dataset.

Cognitive Load Assessment: Recent surveys by Kosch et al. and Giannakakis et al. [14], [26] highlight advances in cognitive load measurement. Notable datasets include WESAD for stress analysis [27] and CLAS for mental state assessment [28]. The Laureate dataset [17] specifically examines cognitive states in educational contexts, but like others, does not integrate

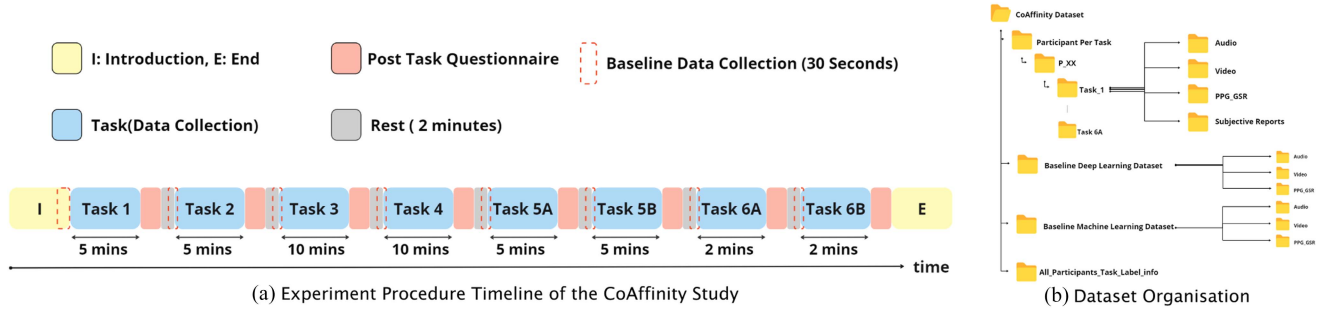


Fig. 2. Figure (a) details the timeline of the experiment, showing the sequence and duration of tasks interspersed with rest periods, baseline data collection, and post-task questionnaires. Figure (b) outlines the organization of the CoAffinity dataset, showcasing the segregation of data by task and modality. It includes audio, video, physiological signals (PPG and GSR), and subjective reports for each participant and task.

both cognitive and affective measurements in collaborative settings.

Remote Collaboration Datasets: The RECOLA dataset [7] represents an early attempt at capturing remote collaboration data, providing 9.5 hours of audio-visual and physiological recordings during dyadic tasks. AVCaffe [18], while groundbreaking in remote work scenarios, has several key limitations: (1) it lacks physiological signals (PPG and GSR) which are essential for objective measurement of cognitive and affective states [29]; (2) several of its tasks focus on individual activities rather than true collaboration, limiting insights into interpersonal dynamics; (3) its fixed task structure doesn't effectively capture the natural progression of emotional states in remote meetings; and (4) it lacks analysis of group synchronization which is crucial for understanding collaborative engagement. CoAffinity addresses these gaps through (1) the incorporation of physiological measurements that provide objective ground truth data alongside self-reported measures; (2) carefully designed collaborative tasks based on empirical pilot studies with 12 participants showing enhanced emotional elicitation; (3) the inclusion of group synchronization metrics (Coherence Mean and Peak) for quantifying interpersonal dynamics; and (4) refined task designs that specifically target collaborative engagement with 2-minute baseline calibration periods between tasks to ensure signal quality and participants returning baseline emotions. Our approach integrates emerging wearable sensor technology with novel analytical methods to investigate physiological synchrony in remote work environments. The CoAffinity dataset fills the following gaps compared to previously discussed literature. First, it addresses the lack of cognitive load and emotion elicitation in naturalistic and realistic remote video conference settings. Second, it captures behavioural (audio, video) and physiological data (PPG and GSR) in these natural settings. Finally, all Audio, Video and Physiological (PPG and GSR) are labelled against cognitive loads via NASA TLX and affective states from the Positive and Negative Affect Schedule (PANAS) and Self-Assessment Manikin (SAM) enhancing the granularity of cognitive and emotional assessments. In summary, The CoAffinity dataset introduces a novel approach by integrating audio, video, and physiological signals to assess cognitive load and emotional responses in a remote collaborative context, specifically within video conferencing scenarios. This integration offers a unique contribution

to the field by providing a comprehensive dataset that captures the subtleties of spontaneous emotional expressions and the mental efforts of participants engaging in real-time collaborative tasks. Unlike datasets focused solely on emotion elicitation through passive viewing of content, CoAffinity's setup involves active, goal-oriented tasks that mimic real-world interactions, providing insights into the interplay between affective states and cognitive processes in a modern collaborative environment. Its comprehensive approach can potentially be a reference for future research on affective states in video conference settings [12], [30].

III. METHODOLOGY

A. Experiment Procedure

The study recruited participants aged 18–60 years fluent in English to ensure effective communication. The experiment was conducted in two designated university rooms, each equipped with laptops, Noise Cancelling Headphones, mice, and Shimmer GSR + PPG Sensors.¹ The setup utilized Zoom for video conferencing and Qualtrics² for questionnaire administration. Participants were assigned unique IDs and allocated to rooms A (odd-numbered) and B (even-numbered). Shimmer devices, worn on the non-dominant hand, recorded GSR from the ring and middle fingers and PPG from the index finger. Prior to the experiment, a pre-study questionnaire captured demographics and interpersonal familiarity, ensuring participants were paired with unfamiliar co-participants to mitigate pre-existing relationship effects.

Each 75–90 minute session consisted of multiple tasks administered through Qualtrics, with participants interacting via Zoom's gallery view. A researcher monitored each session with their audio and video muted, providing technical support only when necessary to maintain data integrity. After each task, participants reported their affective states and cognitive loads, followed by a 2-minute rest period. Fig. 2(a) shows the temporal structure of data collection, including baseline recordings and task sequences.

¹[Online]. Available: <https://shimmersensing.com/product/shimmer3-gsr-unit/>

²[Online]. Available: <https://www.qualtrics.com/au/>

TABLE I
SUMMARY OF EMOTIONAL DATABASES

Database	Participants	Elicitation/Stimuli	Modalities	Annotations
DEAP	32	1-min music clips	EEG, ECG, EMG, GSR, BVP, Respiration, Skin temp.	Valence, Arousal, Dominance, Liking
MAHNOB-HCI	27	5-117s film clips	EEG, ECG, GSR, BVP, Respiration, Skin temp., Face video, Eye tracker	Valence, Arousal, Dominance, Predictability, Emotional keywords
AMIGOS	40	Various videos	Audio, visual, depth, EEG, GSR, ECG	Big-Five personality traits, PANAS, Valence, Arousal, Dominance, Liking, Familiarity, Basic emotions
DECAF	30	1-min music videos, movie clips	ECG, EMG, EOG, MEG, Near-infrared face video	Arousal, valence, dominance
RECOLA	46	Collaborative Task	Visual, Audio, ECG, EDA	Valence, Arousal
K-EmoCon	32	Debates	Audio, Visual, GSR, ECG	Valence, Arousal
AVCaffe	106	Remote work	Audio, Visual	Valence, Arousal, Cognitive load
CoAffinity (Ours)	39	Remote work	Audio, Visual, Text, PPG, GSR	PANAS, Valence, Arousal, Cognitive load

TABLE II
TASKS AND COGNITIVE-AFFECTIVE EFFECTS

Task	Valence	Arousal	Cognitive Load
Open Discussion	High	Low	Low
Lighten the Mood	High	Moderate	Low
Lost at Sea	Moderate	High	Moderate
Tower of Hanoi	Moderate	Moderate	High
Multitask	Low	High	Moderate
Map Task	Low	High	High

The study received approval from the University of Auckland Human Participants Ethics Committee (UAHPEC 27272), with participant consent obtained through secure web-based forms. Participants provided mandatory study participation consent and optional media usage approval, with only approved participant images included in publications.

B. Task Design

Inspired by Sarkar et al. [18], this study employed a structured series of seven tasks designed to simulate remote collaboration scenarios while eliciting varying levels of cognitive load and affect in a controlled experimental setup. The tasks involved dyadic interaction over a video conferencing platform, allowing for the assessment of physiological synchrony in collaborative settings.

The experimental design included an initial phase to establish baseline affective and cognitive states, followed by progressively complex problem-solving tasks that varied in emotional intensity and cognitive demand. Table II summarizes the tasks, their purpose, and their expected cognitive and affective effects.

The Open Discussion task enabled participants to engage in a five-minute casual conversation on non-personal topics (e.g., movies, sports, hobbies) to establish baseline cognitive and emotional states [31]. This was followed by the Lighten the Mood task, where participants shared humorous anecdotes or used pre-provided jokes to modulate emotional state and enhance

group cohesion [32]. Cognitively demanding tasks followed, beginning with Lost at Sea, a survival prioritization exercise that required decision-making under uncertainty and induced moderate cognitive load and emotional engagement [33]. Next, the collaborative Tower of Hanoi assessed strategic thinking and coordination, simulating a co-action task to measure synchrony effects in problem-solving [34]. The final phase introduced high cognitive load tasks. The Multitask Challenge required participants to simultaneously compose an email while solving arithmetic or English-based questions, reflecting common remote work scenarios involving divided attention and high cognitive effort [35]. The experiment concluded with the Montclair Map Task, where dyads collaboratively navigated a map under time constraints, simulating high-pressure decision-making and stress management [36].

Table II details each task, and the expected emotional and cognitive load levels. Detailed task design and purpose is in appendix, available online Section II. These tasks spanned various cognitive loads and affective states relevant to remote work, structured to analyse cognitive and emotional dimensions in collaboration [36], [37]. Task order progressively increased cognitive load, reflecting a typical workday's pressure, starting with low-cognitive-load activities (Open Discussion and Lighten the Mood) to establish rapport and baseline measurements, then progressively increasing in complexity through problem-solving tasks (Lost at Sea, Tower of Hanoi) to high-demand activities (Multitask, Map Tasks), with 2-minute breaks between tasks to prevent fatigue and allow physiological signals to return to baseline. This fixed order and the 2-minute break were deliberately chosen to maintain experimental control and ensure consistent progression of cognitive load across all participants. Although this decision meant that we could not fully account for potential order effects or individual differences in cognitive adaptation, varying the task order might have provided additional insights into how different task sequences impact cognitive load and emotional states in remote collaboration. Cognitive load and emotional data were collected post-task to minimize disruption

during tasks, supporting the effective induction of targeted states and validating task selection for real-world scenarios.

C. Experiment Data Collection Settings

1) *Audio and Video Recording*: Continuous audio-video recordings were made using the Zoom platform to capture detailed behavioural responses. The settings were adjusted to store all audio and video data directly on the local device. This was done to accurately document facial expressions, vocal tone, and other non-verbal cues essential for affective studies. The video data was recorded in MP4 format with a resolution of 960×540 at 25 frames per second for each participant, while the audio was captured in WAV format at a sample rate of 44.1 kHz. To ensure the quality and integrity of the data, participants were instructed to avoid using virtual backgrounds or filters during the sessions.

2) *Physiological Data Collection*: Physiological responses were closely monitored using Shimmer sensors, which captured raw GSR and PPG data. This data was collected at 256 Hz sampling frequency and saved in comma-separated value (CSV) files. Custom software using the lab streaming layer protocol³ for gathering Shimmer data efficiently recorded all sensor outputs, synchronizing them with UNIX timestamps to produce a sequential and organized dataset. These measurements are crucial for analysing physiological indicators of emotional arousal and cognitive load in research studies.

a) *Data Synchronization and Processing*: We developed a custom Python-based pipeline to manage and synchronize data from multiple devices. This pipeline was essential for assigning accurate timestamps to physiological data, aligning it with video and audio recordings and specific task intervals. During data collection, researchers manually entered each task's start and stop times. The software then generated UNIX timestamps for these events and stored them in CSV files. The software also recorded each task session's start and end times in UNIX and real-time formats at the end of each session. This data was later used to segment the audio, video, and physiological data according to the respective task sessions. This timestamping approach ensured precise synchronization of physiological data with corresponding audio-video recordings and self-reported measures.

D. Ground Truths

To accurately assess the cognitive load and affective states of participants, our study used the NASA Task Load Index (NASA-TLX), the Positive and Negative Affect Schedule (PANAS), and Self-Assessment Manikin (SAM) measures.

- NASA-TLX: Measures cognitive load across six subscales, each rated from 0 to 100 in 5-point increments, for detailed cognitive demand analysis [38].
- PANAS: Captures participants' emotions via a 5-point Likert scale [39].
- SAM: Assesses emotional response on arousal and valence using a 9-point scale [40].

The appendix, available online of the paper includes the complete questionnaire used to collect these measures, and the sample responses are visually represented in Fig. 1 to demonstrate typical participant responses. While external annotations for arousal and valence are common in affective computing research [4], [23], [41], [42], we chose self-reported measures. This was for several reasons. First, measuring cognitive load is complex because it includes personal and subjective states like mental demand, effort, and time pressure. Studies show that these internal experiences are hard to accurately assess from external behaviours [23], [43]. Second, emotional ratings can be inconsistent, both within the same person over time (intra-annotator inconsistency) and between different people (inter-annotator inconsistency). This variability, noted in various studies [18], [42], [44], suggests that external annotations may not accurately capture participants affective states. Given these considerations, our study deliberately collected self-reported data to ensure the accuracy and reliability of the measurements related to cognitive load and affective responses. This method avoids the pitfalls associated with external annotations and enhances the integrity of the data, which is crucial for subsequent analyses and conclusions drawn from the study outcomes.

E. Data Processing

Each task and its respective data modalities were treated as a single data point for long sessions. These sessions ranged from 2 to 10 minutes and were labelled based on associated self-report measures. In contrast, short sessions involved segmenting the data from long sessions into 30-second intervals for each task. These segments were non-overlapping and considered independently from participant data, resulting in multiple data points per task per participant. Each segment was labelled according to corresponding self-report measures. We also benchmarked the data from short and long sessions using classical machine learning and deep learning models. Features were extracted across all modalities for the classical models, whereas for the deep learning models, we used cleaned raw data as input. Further details on data processing are provided in subsequent sections. Details of extracted features across all modalities are attached in the Appendix, available online. This methodology for session-based analysis is inspired by the approaches of Sarkar and Laporte et al. [17], [18]. The primary goal of dividing the data into shorter sessions was to increase the number of data points, thus potentially enhancing the analysis's breadth and depth and enabling a comparison of model performances.

1) *Video*: For machine learning (ML) models, video processing involved extracting facial expression features using the OpenFaceFeatureExtraction tool⁴, as detailed by Baltrusaitis et al. [45], in their 2018 study on the OpenFace 2.0 toolkit. Features such as eye gaze direction vectors, head location coordinates, head rotation angles, and the intensity and presence of 17 Action Units (AUs) as defined by the Facial Action Coding System (FACS) were extracted. The mean and standard deviation of

³[Online]. Available: <https://labstreaminglayer.org/>

⁴[Online]. Available: <https://github.com/TadasBaltrusaitis/OpenFace>

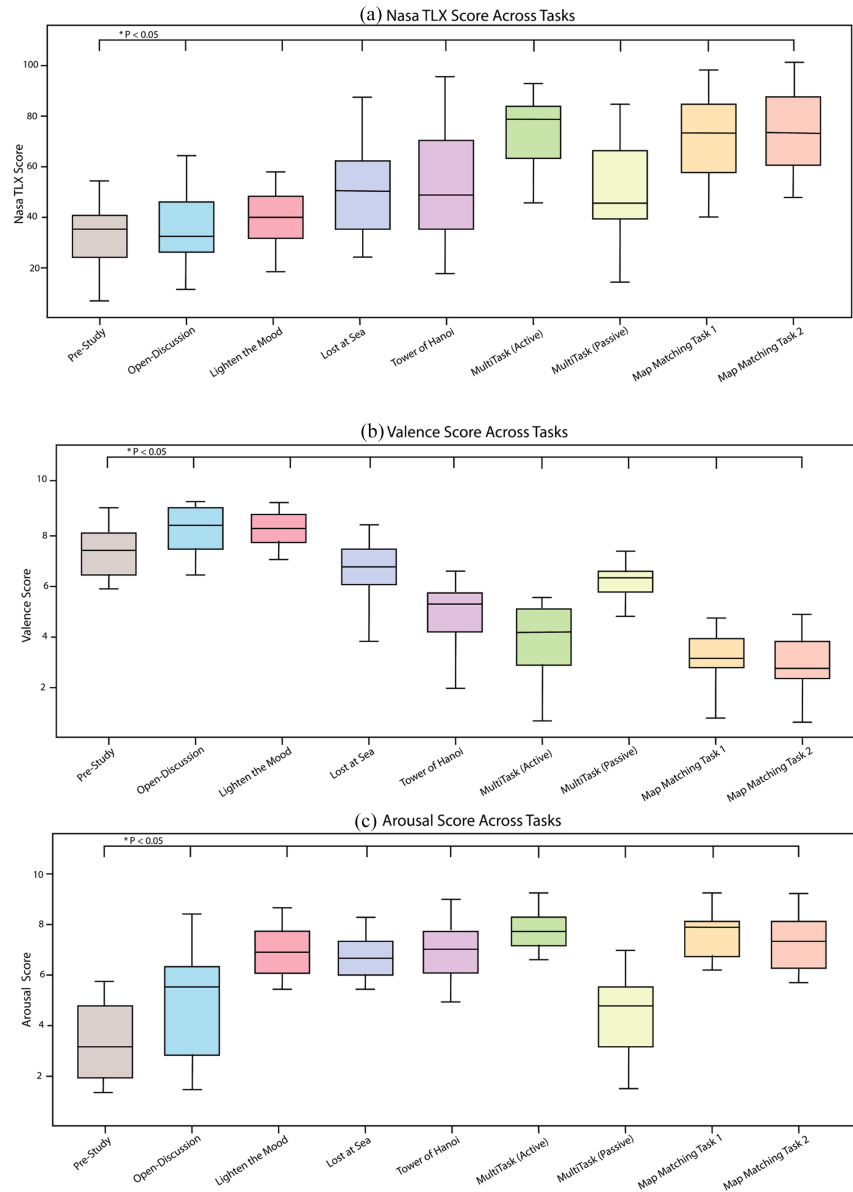


Fig. 3. Distribution of cognitive load (NASA-TLX), Valence and Arousal Scores Across Tasks with Significance Indicators (a) NASA-TLX scores showing cognitive load progression, Box plots revealing the systematic increase from baseline through complex tasks, with peak load in Multitask Active condition.; (b) Valence distribution patterns for each task condition. Higher scores indicate more positive emotional states. Box plots show the progressive modulation of emotional valence, with notable decreases during cognitively demanding tasks.; (c) Arousal levels across different task conditions. Box plots demonstrate the progression from baseline arousal through varying levels of physiological activation. Higher scores indicate increased arousal states. Significant differences between task pairs, as determined by Tukey's HSD test, are indicated by horizontal lines and asterisks (*). The Box plots results validate the CoAffinity dataset's capacity to capture meaningful variations in participant states across different collaborative scenarios.

these features and the mean intensity and presence percentage of AUs were calculated over all frames, resulting in a final set of 52 features that described facial expressions. For deep learning (DL) models, the methodology was adapted from Sarker et al. [18], where long videos were segmented into two-second clips. These clips were then used to train DL models with face crops from each segment, allowing for focused analysis of facial expressions over brief intervals.

2) *Audio*: Similar to video, audio features for ML models were extracted using the OpenSMILE toolkit⁵ with the eGeMAPsv02 feature set, as described by Eyben et al. [46]. This includes a wide range of features such as energy, loudness,

spectral properties, and voice quality measures like jitter and shimmer, amounting to a total of 84 features. These features are essential for assessing vocal expressions related to emotional states.

For DL models, the audio data from long sessions was downsampled to 16 KHz and segmented into two-second clips. Melspectrograms were generated from these segments to serve as inputs for DL models, aligning with the preprocessing techniques used by Sarker et al. [18] for training on concise audio data.

⁵[Online]. Available: <https://audeering.github.io/opensmile/>

3) *Physiological Signal Processing*: The preprocessing of physiological signals (PPG and GSR) followed a systematic approach to ensure data quality and reliability [47]. For PPG signals, we employed a fourth-order Butterworth band-pass filter with cut-off frequencies between 0.5 Hz and 3 Hz to mitigate high-frequency noise from power sources and low-frequency components from capillary actions. The GSR signal processing utilized a second-order Butterworth low-pass filter with a 0.5 Hz cut-off frequency. Both signals underwent dynamic noise reduction through polynomial fitting: third-order for PPG and fifth-order for GSR. To achieve optimal signal smoothing, we applied moving average filters with modality-specific window sizes: 10 samples for PPG and 256 samples for GSR.

Feature extraction was performed using the NeuroKit library [48], with distinct approaches for each signal type. For PPG, we extracted a comprehensive set of heart rate variability (HRV) features, including mean PPG rate, time-domain metrics, frequency-domain components, and nonlinear measurements. The feature set varied by session length: 56 features for short sessions and 68 features for long sessions [49]. GSR processing focused on electrodermal activity (EDA) characteristics, capturing both baseline signals (EDA_Raw) and their components: EDA_Tonic for slower changes and EDA_Phasic for rapid variations in skin conductance. We additionally extracted detailed skin conductance response metrics, including onset points, peak values, height measurements (onset to peak distance), amplitude magnitudes, rise times (onset to peak duration), and recovery periods (time to half-amplitude return). These GSR features, totalling 36, provide a thorough characterization of electrodermal responses [48].

For deep learning applications, both signals underwent consistent preprocessing followed by uniform segmentation into two-second intervals, following a methodology similar to Kang et al. [47]. This approach ensured temporal alignment with audio and video modalities, facilitating synchronized multimodal analysis [18]. The cleaned signals served as direct inputs to the deep learning models, maintaining the temporal relationships crucial for integrated analysis across all modalities.

F. Dataset Organisation and Statistics

The CoAffinity dataset contains multimodal recordings from 39 participants (17 females, 22 males, mean age 21.8 ± 3.11 years), collected over 38 hours of controlled task conditions. The final participant count reflects the exclusion of one participant's data due to a saving error in the paired study design. The dataset comprises 1,248 data points, capturing interactions between affective states and physiological signals across various collaborative tasks.

The dataset follows a hierarchical organization structure to ensure accessibility and research utility. Each participant's data (IDs 1-39) is organized into eight task-specific subfolders, with each task folder containing four modality-specific directories: audio, video, GSR, and PPG. The physiological data folders (PPG and GSR) include 30-second baseline recordings for calibration purposes. Supplementary data files include participant

TABLE III
DATASET CONTENT SUMMARY

Parameter	Details
Subjects	39
Duration	38 hrs
# Data Points	1248
Video	960 X 540 at 25fps
Audio	44.1KHz
PPG	256 HZ
GSR	256 HZ
Gender	Male: 22, Female: 17
Age	18 to 24 : 10 25 to 34 : 27 34 to 40 : 2
Ground truths / Rating scales	Arousal, Valence, PANAS , Mental Demand, Temporal Demand, Effort Physical Demand, Performance, and Frustration
Session Data Size	Short: 3432 datapoints, Long: 312 datapoints for each modality.

pairing information in CSV format and task-specific subjective ratings for each session.

Table III provides a comprehensive overview of the dataset statistics, while Fig. 2(b) illustrates the organizational structure. This systematic organization facilitates efficient data access and analysis for affective computing research, with clear delineation between modalities and tasks maintaining data integrity and usability.

IV. ANALYSIS AND EVALUATION

We utilized Analysis of Variance (ANOVA) to assess differences in the NASA-TLX, SAM and PANAS scores across tasks, followed by Tukey's Honestly Significant Difference (HSD) test for pairwise comparisons [50]. These methods helped identify significant differences in perceived cognitive load and emotional responses across various task settings. Detailed results of ANOVA and Tukey's pairwise comparisons, including significant differences between specific task pairs, are presented in Appendix, available online Section 3, providing a comprehensive understanding of cognitive demand and emotional response variations across tasks.

A. NASA-TLX Cognitive Load Analysis

The analysis of NASA-TLX scores revealed systematic variations in cognitive load across different tasks, demonstrating the effectiveness of our progressive task design, as shown in Fig. 3(a). ANOVA revealed significant differences in cognitive load across tasks ($F = 34.22$, $p < 0.001$), indicating a substantial variation in how demanding each task was perceived by participants. Pairwise comparisons using Tukey's Honestly Significant Difference (HSD) test highlighted specific tasks that significantly differed from each other. For example, Tasks 3, 4, 5A, and 6B were found to be significantly more demanding compared to the Pre-Study task. The pre-study task, designed to establish baseline cognitive load, exhibited low mean NASA-TLX scores ($M = 33.03$, $SD = 12.94$), indicating minimal cognitive demand. The *Open Discussions* and *Lighten the Mood* tasks, which aimed to engage participants without imposing high

cognitive load, showed similar scores (*Open Discussions*: $M = 35.50$, $SD = 13.58$; *Lighten the Mood*: $M = 37.47$, $SD = 9.64$). No significant differences were observed between these initial tasks and the pre-study baseline ($p > 0.05$), suggesting that participants experienced consistent cognitive load levels during the early stages of the study.

Cognitive load significantly increased in the *Lost at Sea* task ($M = 56.42$, $SD = 15.89$) and the *Tower of Hanoi* task ($M = 56.49$, $SD = 18.78$), both of which required active problem-solving and collaboration. Pairwise comparisons confirmed that these tasks had significantly higher cognitive load compared to the initial tasks ($p < 0.001$), indicating the intended escalation in cognitive demand. The *Multitask Active* task showed the highest NASA-TLX score ($M = 70.50$, $SD = 13.42$), significantly surpassing all previous tasks ($p < 0.001$), reflecting substantial cognitive strain due to the complexity and time constraints of the task. In contrast, the *Multitask Passive* task showed a notable reduction in cognitive load ($M = 41.26$, $SD = 14.99$), significantly lower than *Multitask Active* ($M = 70.50$, $SD = 13.42$) ($p < 0.001$). This decrease suggests that the task, while still engaging, imposed a lower mental and temporal demand, possibly due to its passive nature. The *Map Matching Tasks A* and *B* maintained relatively high cognitive load scores (*Map Matching Task A*: $M = 60.56$, $SD = 14.72$; *Map Matching Task B*: $M = 61.13$, $SD = 14.60$). While these scores were higher than those observed during the initial tasks ($p < 0.001$), they were not significantly different from the scores for *Lost at Sea* and *Tower of Hanoi*, indicating a sustained level of cognitive effort towards the end of the study.

Tukey's HSD test identified key differences between tasks. *Multitask Active*, the most demanding task, differed significantly from all other tasks except the *Map Matching Tasks A* and *B*, where cognitive load remained elevated. These results suggest that tasks involving multitasking or complex problem-solving under time constraints elicit the highest cognitive load, while simpler tasks such as *Open Discussions* induce lower cognitive demand.

The progressive increase in NASA-TLX scores aligns with the theoretical expectations of cognitive load theory, demonstrating that increased task complexity and demand elevate mental and temporal pressure. These findings underscore the importance of task design in managing cognitive load, particularly in remote or collaborative settings.

B. Valence and Arousal Analysis

The self-reported Valence and Arousal assessments showed significant variations across tasks, as confirmed by ANOVA results: Valence ($F = 82.75$, $p < 0.0001$) and Arousal ($F = 45.22$, $p < 0.0001$). The low p-values suggest that participants experienced distinct emotional and physiological responses depending on the nature of each task. Pairwise comparisons using Tukey's HSD test highlighted specific tasks that significantly differed from each other. For example, Significant differences in Valence were observed particularly between high-valence tasks (e.g., Task 1 and Task 2) and tasks associated with negative emotions (e.g., Task 4 and Task 5A). In addition, Arousal ratings showed that tasks designed to be more engaging or stressful (e.g.,

Task 2 and Task 6B) resulted in higher arousal levels compared to baseline or less demanding tasks.

The mean Valence scores across tasks ranged from 3.08 to 7.77, indicating a spectrum of emotional responses from low valence (negative/low pleasure) to high valence (positive/high pleasure). The pre-study task exhibited moderate valence levels ($M = 6.95$, $SD = 1.47$), consistent with the fact that participants were becoming familiarized with the experiment, which likely did not evoke strong emotional responses. The *Open Discussion* task showed a relatively high mean valence score ($M = 7.64$, $SD = 1.16$). This task involved free-form conversation and exchange of ideas, which participants found engaging and enjoyable. The *Lighten the Mood* task exhibited a similarly high valence score ($M = 7.77$, $SD = 1.01$), reflecting a positive emotional response, possibly due to the nature of the task, which aimed to induce a light-hearted and relaxed atmosphere. Tukey's pairwise comparisons revealed no significant differences between these two tasks and the pre-study task ($p > 0.05$), suggesting similar positive emotional experiences during these stages. In contrast, tasks like *Tower of Hanoi* and *Multitask Active* exhibited substantially lower valence scores ($M = 3.82$, $SD = 1.62$; $M = 3.08$, $SD = 1.29$, respectively). These tasks, characterized by high cognitive demand and complexity, likely led to increased frustration or stress, reflected in their significantly lower valence ratings. Tukey's HSD test identified significant differences between these challenging tasks and the more positively rated tasks like *Open Discussion* and *Lighten the Mood* ($p < 0.001$), as shown in Fig. 3(b). This suggests that cognitively intensive tasks had a notable negative impact on participants' emotional states.

Arousal scores also varied significantly across tasks, with means ranging from 3.26 to 7.85. The pre-study task showed low arousal ($M = 3.26$, $SD = 1.62$). The *Open Discussion* task maintained a relatively low arousal level ($M = 4.44$, $SD = 2.05$), indicating that the task was engaging but did not induce high physiological excitement or stress. Higher arousal levels were observed in more demanding tasks, such as *Multitask Active* ($M = 7.85$, $SD = 1.35$) and *Map Matching Task A* ($M = 7.18$, $SD = 1.14$). These tasks required participants to perform under time constraints or handle multiple tasks simultaneously, contributing to heightened arousal levels. Tukey's HSD test revealed significant differences between these high-arousal tasks and tasks such as the pre-study and *Open Discussion* ($p < 0.001$), indicating that these tasks imposed higher physiological demands. The *Lost at Sea* and *Tower of Hanoi* tasks, which also involved substantial cognitive effort, showed moderately high arousal ($M = 7.00$, $SD = 0.76$; $M = 6.87$, $SD = 1.38$, respectively). This aligns with the theory that complex tasks requiring significant problem-solving elicit greater physiological activation as participants are required to engage deeply with the task. The Tukey HSD test revealed several important pairwise differences in both Valence and Arousal across tasks, as shown in Fig. 3(c). For Valence, tasks that induced lower pleasure, such as *Tower of Hanoi* and *Multitask Active*, were significantly different from tasks with higher pleasure, like *Open Discussion* and *Lighten the Mood* ($p < 0.001$). Similarly, for Arousal, tasks with low physiological activation, such as pre-study and *Open Discussion*, were significantly different from tasks that elicited high activation,

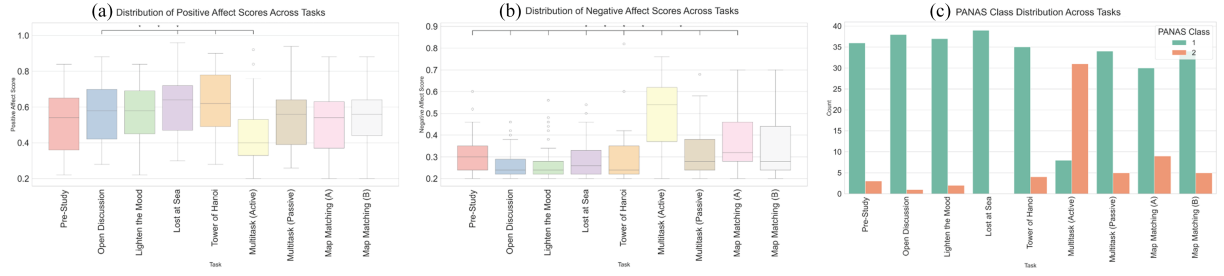


Fig. 4. Distribution of Positive and Negative Affect Scores (PANAS) across tasks. (a) Positive affect varies by task type. (b) Negative affect shows distinct patterns across tasks. (c) PANAS class distribution highlights differences in affective states per task.

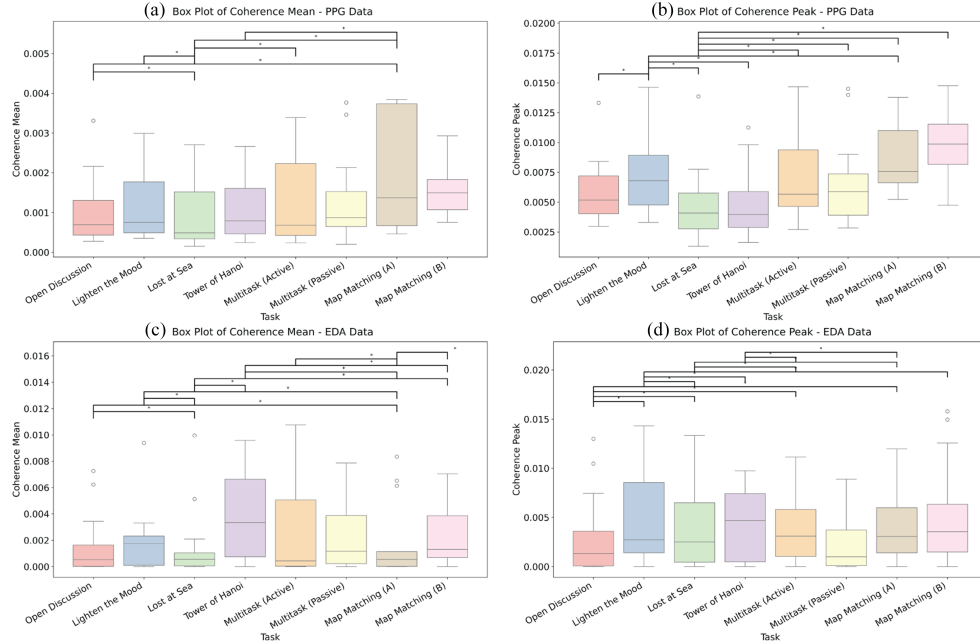


Fig. 5. Box plots of **Coherence Mean** and **Coherence Peak** across different tasks for PPG and EDA data. Each subplot represents coherence values across the defined tasks: *Open Discussion*, *Lighten the Mood*, *Lost at Sea*, *Tower of Hanoi*, *Multitask (Active)*, *Multitask (Passive)*, *Map Matching (A)*, and *Map Matching (B)*. (a) PPG Coherence Mean, (b) PPG Coherence Peak, (c) EDA Coherence Mean, and (d) EDA Coherence Peak. Significance markers (*) indicate significant differences in coherence values based on Wilcoxon pairwise comparisons ($p < 0.05$). Tasks with higher social engagement (e.g., *Open Discussion*, *Lighten the Mood*) tend to exhibit greater coherence, whereas cognitively or spatially demanding tasks (e.g., *Map Matching*) show reduced coherence.

such as *Multitask Active* and *Map Matching Task A* ($p < 0.001$). These results highlight distinct emotional profiles of each task, demonstrating how task demands shape emotional valence and arousal profiles.

Overall, the Valence and Arousal ratings provide valuable insights into participants' emotional and physiological responses across different tasks. The results suggest that tasks involving higher cognitive load or multitasking, such as *Multitask Active* and *Map Matching Task A*, not only increased physiological arousal but also reduced subjective pleasure, as indicated by the lower valence ratings. In contrast, tasks with lower demands, such as *Open Discussion* and *Lighten the Mood*, elicited higher valence and lower arousal, reflecting a more positive and relaxed participant experience.

C. PANAS Analysis

Our study investigated the affective responses associated with various tasks using the PANAS questionnaire. The results, as depicted in the accompanying Fig. 4, provide insight into

the emotional states of participants in the context of different cognitive demands. The self-reported Positive affect and Negative affect showed significant variations across tasks, as confirmed by ANOVA results: Positive affect ($F = 3.70, p < 0.001$) and Negative ($F = 16.1845, p < 0.0001$).

Fig. 4(a) illustrates the distribution of Positive Affect scores across tasks. Post-hoc tests revealed that Positive Affect remained stable, with no substantial decreases except for *Multitask (Active)* ($p = 0.0419, M = 0.61, SD = 0.16$), *Tower of Hanoi* ($p = 0.0003, M = 0.63, SD = 0.17$), and *Lost at Sea* ($p = 0.0008, M = 0.62, SD = 0.17$), which showed significantly lower scores compared to *Open Discussion* ($M = 0.57, SD = 0.17$). The pre-study phase exhibited a higher mean score ($M = 0.53, SD = 0.18$), likely reflecting an initial neutral or anticipatory emotional state. However, most tasks did not significantly differ from each other, suggesting that cognitive load variations had minimal impact on positive emotions.

Conversely, Fig. 4(b) demonstrates the Negative Affect scores, where a notable variation is seen. The Pre-Study phase had the lowest mean Negative Affect score ($M = 0.30, SD =$

0.09), serving as a baseline. The *Multitask Active* condition ($p < 0.001$, $M = 0.38$, $SD = 0.11$) elicited the highest Negative Affect, significantly higher than *Open Discussion* ($p < 0.001$, $M = 0.26$, $SD = 0.07$), *Lighten the Mood* ($p < 0.001$, $M = 0.27$, $SD = 0.08$), and *Lost at Sea* ($p < 0.001$, $M = 0.29$, $SD = 0.09$). This suggests that *Multitask Active* induced greater emotional strain, likely due to the simultaneous execution of multiple cognitive tasks. Similarly, the *Montclair Map Tasks* showed significantly elevated Negative Affect compared to the Pre-Study condition ($p = 0.005$, $M = 0.35$, $SD = 0.10$). Post-hoc analysis also identified significant increases in Negative Affect for *Multitask Active* compared to *Tower of Hanoi* ($p < 0.001$, $M = 0.32$, $SD = 0.09$) and *Multitask Passive* ($p < 0.001$, $M = 0.31$, $SD = 0.09$), reinforcing the idea that active multitasking contributes to increased frustration and cognitive overload. In contrast, *Lighten the Mood* ($p < 0.001$, $M = 0.27$, $SD = 0.08$) and *Open Discussion* ($p < 0.001$, $M = 0.26$, $SD = 0.07$) consistently exhibited the lowest Negative Affect scores, suggesting that these tasks provided a stress-relieving effect, as indicated by significantly lower levels of reported distress compared to higher-demand tasks.

Fig. 4(c) presents the PANAS class distribution across tasks, delineating between classes 1 and 2. Class 1, representing a more positive affective state, dominates the distribution in almost all tasks. This dominance is particularly evident in the *Open Discussion* ($M = 0.57$, $SD = 0.17$) and *Lighten the Mood* ($M = 0.58$, $SD = 0.16$) tasks, underscoring their efficacy in fostering a positive environment. Conversely, class 2 shows a notable increase during the *Multi-Task (Active)* ($M = 0.38$, $SD = 0.11$) and *Montclair Map Tasks* ($M = 0.35$, $SD = 0.10$), indicating a shift towards a more negative affective state. This shift could reflect the stress or cognitive strain induced by the tasks.

Overall, the data suggest a complex interaction between task demands and affective responses. While the positive affect appears relatively stable, the negative affect is more sensitive to the nature of the task. Tasks perceived as less demanding or more engaging might preserve or enhance positive affect, while challenging tasks seem to elicit a stronger negative response. This relationship underscores the importance of task design in educational and occupational settings, where balancing cognitive demands with emotional well-being can be crucial for performance and satisfaction.

D. Inter Label Relationship

In our study, we explored the interrelationships between cognitive load and affective state using the NASA-TLX, SAM, and PANAS scales across various tasks. We performed the Pearson Correlation test using normalized self-reported scores. The overall correlation matrix for each task is provided in Appendix, available online Section 4.

We observed that mental demand was positively correlated with both physical and temporal demand. However, the strength of this relationship varied by task. For instance, in the *Tower of Hanoi* task, known for its cognitive intensity, the correlation between mental and physical demand was notably strong ($r = 0.49$, $p < 0.05$), reinforcing the theory that mentally taxing

tasks can elevate perceptions of physical effort and temporal urgency. Additionally, mental demand was positively correlated with temporal demand in *Lost at Sea* ($r = 0.56$, $p < 0.01$), emphasizing the link between cognitive and time pressure.

Performance exhibited varied correlations with NASA-TLX dimensions, showing a significant negative relationship with effort, particularly in tasks such as *Lost at Sea* ($r = -0.48$, $p < 0.05$) and *Map Matching* task ($r = -0.56$, $p < 0.01$). This suggests that the immersive nature of the task might increase the perceived effort, leading to lower performance outcomes. Similarly, frustration had a significantly negative correlation with performance, most evident in the *Map Matching* task ($r = -0.62$, $p < 0.01$) and multitask ($r = -0.47$, $p < 0.05$), implying that navigational challenges and multitasking could induce frustration, hampering task performance.

Affective states revealed strong connections with workload dimensions. Negative affect exhibited a robust positive correlation with effort and frustration across several tasks. This trend was particularly higher in the *Multi-Task (Active)* condition, where frustration was highly correlated with negative affect ($r = 0.60$, $p < 0.01$), and effort was similarly correlated ($r = 0.46$, $p < 0.05$). This is likely due to the demanding nature of the task, eliciting stronger negative emotions. In the *Lost at Sea* task, frustration showed a high correlation with negative affect ($r = 0.41$, $p < 0.05$), supporting this relationship.

Conversely, positive affect was positively correlated with performance, with the open discussion phase showing a significant association ($r = 0.46$, $p < 0.05$) and *Lighten the Mood* ($r = 0.45$, $p < 0.05$), suggesting that initial engagement with study materials and positive interventions may induce a positive mood, promoting better performance outcomes. Notably, the correlation profiles were not uniform across tasks, indicating task-specific dynamics in the interplay between workload, affect, and performance. The *Open Discussion* and *Lighten the Mood* tasks, for instance, demonstrated unique correlation patterns that need further exploration. The *Open Discussion* task showed a moderate correlation between effort and frustration ($r = 0.52$, $p < 0.05$), indicating that discussions requiring cognitive effort may also induce frustration. The *Map Matching* task displayed a significant correlation between frustration and negative affect ($r = 0.63$, $p < 0.01$), reflecting the cognitive challenge inherent in the task. These findings emphasize the importance of considering task context in understanding subjective workload and affect, highlighting how different tasks shape cognitive load and emotional response [13].

E. Group Synchronization

We examined group synchrony via Mean and Peak Coherence in PPG and EDA signals to reveal task-specific differences [20], [51]. Mean Coherence measures sustained frequency-domain alignment, while Peak Coherence highlights transient, high-synchrony events. We hypothesized that cognitively demanding, less socially interactive tasks would show lower synchrony than less demanding, socially engaging ones.

a) *Mean Coherence*: Given the data's non-normal distribution (via Shapiro–Wilk), a Friedman test on PPG data

indicated a statistically significant main effect for Mean Coherence in PPG signals across the eight tasks ($p < 0.0001$). Wilcoxon pairwise comparisons revealed that *Open Discussion* and *Lighten the Mood* both exhibited substantially higher mean coherence levels than tasks such as *Lost at Sea*, *Map Matching (A)*, and *Multitask (Active)*. Specifically, comparing *Open Discussion* to *Lost at Sea* yielded $p = 0.0054$, while *Lighten the Mood* versus *Lost at Sea* resulted in $p = 0.0007$. Likewise, shifting from *Open Discussion* to *Map Matching (A)* showed a significant reduction ($p = 0.0012$), underscoring the impact of spatially intensive tasks on disrupting cardiovascular synchrony. Notably, *Lost at Sea* also differed significantly from *Tower of Hanoi*, *Multitask (Active)*, *Multitask (Passive)*, *Map Matching (A)*, and *Map Matching (B)*, highlighting its influence on group synchrony once decision-making task is introduced. In general, *Map Matching (A)* demonstrated multiple steep drops in coherence when compared to less complex tasks, reflecting the pronounced effect of high-load, spatially oriented scenarios on PPG synchrony.

Results suggest that PPG mean coherence remains higher for socially interactive or low-demand activities, such as *Open Discussion* or *Lighten the Mood*, but shows pronounced declines when participants engage in parallel or complex cognitive tasks (e.g., *Map Matching (A)*, *Multitask (Active)*). Tasks involving casual conversation or humour sustain greater group-level synchrony [20]. These findings imply that shared focus fosters PPG coherence, whereas distributed attention or complex manipulation tasks (e.g., *Map Matching (A)*) disrupt synchronised engagement.

Friedman test for EDA mean coherence yielded similar overall significance ($p < 0.0001$). Wilcoxon analyses indicated that socially driven contexts, such as *Open Discussion* and *Lighten the Mood*, maintained higher electrodermal synchrony when compared to more intricate or spatial tasks. For instance, *Open Discussion* to *Lost at Sea* showed a significant drop ($p = 0.0054$), and *Lighten the Mood* to *Lost at Sea* was similarly lowered ($p = 0.00073$). Comparing *Open Discussion* to *Map Matching (A)* ($p = 0.0012$) further underscored the negative impact of spatial complexity on EDA coherence. *Lost at Sea* also displayed multiple differences (e.g., vs. *Tower of Hanoi*, *Multitask (Active)*, *Map Matching (B)*), illustrating how strategic problem-solving modifies autonomic patterns. Overall, *Multitask (Active)* and *Map Matching (A)* consistently elicited deeper declines in EDA coherence than socially engaged tasks.

The decline in mean coherence under cognitively or spatially challenging tasks reflects participants' divergent arousal levels. When tasks require collective emotional engagement (e.g., *Lighten the Mood*), EDA aligns more uniformly across group members, whereas tasks that concentrate on individual problem-solving or resource allocation lead to decreased synchrony in skin conductance responses. EDA-based findings emphasize how social contexts promote higher group synchrony, likely due to shared affective cues. As tasks become more mentally demanding (e.g., *Multitask (Active)*, *Map Matching (A)*), individuals' arousal patterns diverge. These patterns suggest that group EDA coherence depends on shared emotional or cognitive states, with disruptions occurring due to individual stress responses.

When contrasting PPG and EDA for mean coherence, both signals display strong coherence in less demanding or socially rich tasks, with significant declines in high-load or spatial tasks. PPG often exhibits steep transitions (e.g., from high coherence in *Lighten the Mood* to low coherence in *Map Matching (A)*), while EDA can show a sharper drop under stressful conditions (e.g., *Multitask (Active)*). Overall, both multitasking and spatial tasks lead to a decrease in mean coherence across both physiological signals. However, they show different sensitivities: PPG coherence tends to respond quickly to changes in social engagement, while EDA coherence reflects longer-lasting effects related to arousal and stress.

b) Peak Coherence: For peak coherence in PPG, the Friedman test again indicated a significant main effect ($p < 0.001$). Wilcoxon comparisons showed that *Lighten the Mood* yielded notably higher peak coherence than *Open Discussion* ($p = 0.0130$), suggesting humour heightens cardiovascular synchrony more than general social conversation. Moreover, transitions from *Lighten the Mood* to tasks such as *Map Matching (A)* ($p < 0.001$) displayed complete disruption of short-term cardiovascular synchrony, exemplifying how abrupt shifts to spatial complexity can fragment synchrony. Similarly, *Lost at Sea*, *Tower of Hanoi*, and *Multitask (Active)* each elicited distinct decreases in peak coherence when compared with socially oriented tasks, reinforcing that momentary bursts of shared cardiovascular engagement are difficult to sustain under high cognitive load. Peak coherence in heart-rate activity corresponds to transient yet intense periods of synchrony, often tied to emotional contagion or moments of collective excitement. Tasks emphasizing fun and social bonding (*Lighten the Mood*) can spark these short-lived spikes. However, once a task demands fast-paced problem-solving (*Map Matching (A)*, *Multitask (Active)*), peak coherence dissipates.

EDA peak coherence revealed a closely mirrored pattern. While *Open Discussion* and *Lighten the Mood* foster intermittent but robust electrodermal synchrony, pivoting to tasks like *Multitask (Active)* or *Map Matching (A)* extinguishes these peaks. In some comparisons (e.g., *Lost at Sea* vs *Map Matching (A)*), the $p < 0.001$ indicated groups arousal responses were unconnected and independent from each other. Nevertheless, *Tower of Hanoi* and *Lost at Sea* maintain moderate peak coherence, suggesting that tasks involving team-based strategy still allow short bursts of collective physiological reactivity, albeit less so than purely social contexts. Peak electrodermal coherence typically reflects simultaneous, short-term arousal elevations. While humour or casual tasks can produce these collective spikes, high-load or puzzles impose more individualized stress, scattering arousal peaks among participants. Such findings emphasize EDA's role in capturing discrete emotional surges within a group, as opposed to the continuous, collective engagement more commonly observed in social or motivational tasks. When comparing PPG and EDA peak coherence, both signals highlight that brief moments of physiological synchrony are more common in light-hearted or socially engaging tasks but tend to disappear in multitasking or complex problem-solving activities. PPG often shows sharp and rapid changes in peak coherence, suggesting sudden increases in shared excitement or relaxation. In contrast, EDA

can remain partially synchronized when participants experience similar levels of stress. However, in highly demanding tasks such as Map Matching (A), differences in sympathetic arousal become more pronounced, leading to greater divergence in EDA responses. Thus, each measure captures distinct aspects of group emotional synchrony: PPG reflects immediate cardiovascular synchrony, while EDA indicates shared or differing longer-term stress responses.

c) Summary: Across both mean and peak coherence measures, PPG and EDA converge on the conclusion that task structure, whether a casual conversation, humour-based exchange, or cognitively intensive tasks, strongly shapes group-level physiological synchrony. *Lighten the Mood* and *Open Discussion* reliably foster higher coherence, reflecting a social or emotionally open environment conducive to collective affective states. In contrast, tasks that require individual focus, spatial processing, or multitasking severely reduce coherence, highlighting the fragmentation of group emotional or arousal patterns under high workloads. These findings on group synchrony can be used to predict emotional contagion, allowing for the forecasting of an individual group member's emotional state and cognitive load. This approach aligns with the work of Bota et al. [51], who demonstrated how physiological synchrony can serve as an indicator of shared emotional and cognitive experiences within a group.

F. Training Labels

For training our machine learning and deep learning models, we implemented a multi-class classification system for both cognitive load and affect attributes.

1) Cognitive Load Classification: Cognitive load was quantified using the weighted NASA Task Load Index (NASA-TLX) score, which amalgamates ratings across six subscales: Mental Demand, Physical Demand, Temporal Demand, Performance, Effort, and Frustration. The final scores are computed by combining the intensity ratings on these subscales with weights derived from pairwise comparisons [52]. We categorized cognitive load into three distinct levels based on score ranges. *Low cognitive load (Class 1)* corresponds to scores between 1 and 33, indicating minimal mental effort. *Medium cognitive load (Class 2)* encompasses scores from 33 to 66, reflecting a moderate level of cognitive demand. *High cognitive load (Class 3)* includes scores ranging from 66 to 100, representing substantial cognitive effort. This classification provides a structured framework for analyzing variations in cognitive load across different tasks and conditions.

2) Affect Attributes Classification: For affect attributes, we utilized the Self-Assessment Manikin (SAM) dimensions, grounded in Russell's circumplex model of affect [53]. This model represents the effect in a two-dimensional space defined by Valence and Arousal. We classified these dimensions separately, categorizing each into three distinct levels. *Valence* was classified as low for scores ranging from 1 to 3, medium for scores between 3 and 6, and high for scores from 6 to 9. Similarly, *arousal* was categorized as low for scores between 1 and 3, medium for scores from 3 to 6, and high for scores ranging

TABLE IV
CLASS DISTRIBUTIONS FOR COGNITIVE LOAD, VALENCE, AND AROUSAL

Class	Cognitive Load	Valence	Arousal
1	55	84	25
2	167	110	109
3	90	118	178

from 6 to 9. This classification framework provides a structured approach to analyzing variations in emotional and physiological responses across different conditions.

The classification represents the emotional spectrum, with low valence indicating negative or unpleasant emotions, medium valence reflecting neutral or mixed emotions, and high valence denoting positive or pleasant feelings. Arousal classification measures the intensity of emotional energy, where low arousal indicates calmness, medium arousal represents moderate energy, and high arousal signifies excitement or heightened activity. This classification scheme results in a 3×3 grid, yielding nine distinct affective states, allowing for a more comprehensive analysis of emotional responses in remote collaborative settings [54].

As illustrated in Table IV, the distribution of data points across short and long sessions is relatively balanced for cognitive load classes. However, we observed an imbalance in the distribution of affect attribute classes, particularly in the low valence and low arousal categories. This imbalance is primarily attributed to limitations in our study design, which did not effectively elicit states associated with low valence and low arousal in the remote collaborative context.

V. TRAINING AND EVALUATION

In this section, we present machine learning (ML) and deep learning baselines on CoAffinity in both uni-modal and multi-modal setups. Our approach combines traditional ML methods and current deep learning techniques, a decision driven by two key factors: the relatively small number of participants (39) and the complex nature of physiological signals. ML models often excel with smaller datasets and can effectively handle the high-dimensional, non-linear characteristics of physiological data while maintaining result interpretability [55]. To complement this, we included additional dataset evaluation techniques that highlight the importance of physiological signals in assessing cognitive load and affect, particularly in multimodal deep-learning setups. This comprehensive approach allows us to leverage the strengths of both ML and deep learning paradigms, ensuring robust analysis despite the dataset's size limitations while also exploring the full potential of integrating physiological data with audio-visual information for more accurate cognitive and affective state prediction in remote collaborative settings. Detailed performance metrics, including precision-recall characteristics and per-class analysis, are provided in Appendix, available online Section IV.

A. Machine Learning Models

Training: The preprocessing phase addressed two fundamental challenges in multimodal affect recognition: high feature

TABLE V
OPTIMIZED FEATURE DISTRIBUTION ACROSS MODALITIES AFTER
CORRELATION-BASED SELECTION

Modality	Short Sessions	Long Sessions	Feature Categories
Audio	40	44	Spectral, prosodic, voice quality
Video	36	36	Facial AUs, head pose, expressions
PPG	40	55	HRV metrics, frequency and nonlinear components
GSR	40	41	Phasic/tonic components, metrics

dimensionality and limited sample size. Following established methodologies in physiological signal processing [51], we employed correlation-based feature selection with a threshold of 0.85 to eliminate redundant features while preserving essential signal characteristics. This dimensionality reduction process yielded optimized feature sets for both short (30-second) and long (2-10 minute) sessions, as in Table V.

Our experimental framework implemented two distinct approaches: (1) unimodal analysis utilizing modality-specific handcrafted features and (2) multimodal analysis employing early fusion strategies to integrate features across modalities prior to model training. Following the approaches surveyed by Saganowski et al. [13], we focused on three high-performing classifiers: XGBoost, Gradient Boost, and Extra Trees, based on preliminary evaluations. Performance metrics for additional models, including Support Vector Machines (SVM), AdaBoost, Random Forest, k-Nearest Neighbors (kNN), and Naive Bayes, are documented in Appendix, available online Section IV. A comprehensive enumeration of extracted features for each modality is presented in Appendix, available online Section III.

Evaluation Protocol: To ensure robust generalization despite the limited dataset size ($N = 39$ participants), we implemented a Leave-Four-Pair-Out (LFPO) cross-validation strategy, adapting the traditional Leave-One-Subject-Out approach. This methodology specifically addresses the challenges of interpersonal variability in affect recognition tasks. In each iteration, four participant pairs (8 participants) were systematically held out for testing while the model was trained on the remaining data ($N-8$). This process was repeated iteratively across all possible participant pairs, ensuring that every pair was included in the test set at least once while maintaining strict participant independence between training and testing sets. This approach is particularly crucial given the dyadic nature of the dataset, where interactions between participants influence cognitive load and affective states.

Model optimization was conducted through grid search within each LOPO fold, employing the weighted F1-score as the primary evaluation metric. This choice was particularly important given the inherent class imbalance in affect labels. The optimization process, implemented using Scikit-Learn [56], required approximately 124 hours on an AMD 16-core processor to complete.

1) Machine Learning Results: Following our LFPO cross-validation protocol across 39 participants (19 pairs plus one individual), we evaluated model performance using weighted F1

scores to account for class imbalance [5], [18]. The training was conducted using the optimized feature sets described in Table V, with results averaged across all LOPO folds. The consolidated results across all model configurations are presented in Table VI, with detailed performance metrics, confusion matrices, and statistical analyses, and permutation testing of model performance in Appendix, available online Section IV.

Unimodal Performance Analysis: Analysis of individual modalities revealed systematic performance patterns across the cognitive load and affect recognition tasks. Visual modality, implemented through Gradient Boosting, consistently demonstrated superior unimodal performance ($M = 63.0\%$, $SD = 1.8\%$ for short sessions; $M = 59.5\%$, $SD = 1.5\%$ for long sessions in cognitive load recognition). This modality maintained its lead across valence ($M = 60.5\%$, $SD = 1.6\%$ short; $M = 57.0\%$, $SD = 1.4\%$ long) and arousal recognition ($M = 59.5\%$, $SD = 1.7\%$ short; $M = 56.0\%$, $SD = 1.3\%$ long), highlighting the robust discriminative power of visual features in affect recognition.

Audio features, optimally processed through Gradient Boosting, achieved the second-best unimodal performance (cognitive load: $M = 62.0\%$, $SD = 1.6\%$ short, $M = 57.5\%$, $SD = 1.4\%$ long; valence: $M = 58.5\%$, $SD = 1.5\%$ short, $M = 55.0\%$, $SD = 1.3\%$ long; arousal: $M = 58.0\%$, $SD = 1.5\%$ short, $M = 53.5\%$, $SD = 1.2\%$ long). Notably, XGBoost showed comparable performance (within 0.5% difference), suggesting the robustness of audio feature extraction.

Physiological signals, while providing unique insights into participants' internal states, showed relatively lower but still significant performance when processed through XGBoost (cognitive load: $M = 60.0\%$, $SD = 1.4\%$ short, $M = 55.5\%$, $SD = 1.2\%$ long; valence: $M = 57.0\%$, $SD = 1.3\%$ short, $M = 53.5\%$, $SD = 1.1\%$ long; arousal: $M = 56.5\%$, $SD = 1.3\%$ short, $M = 52.0\%$, $SD = 1.1\%$ long).

Multimodal Integration Performance: The optimal trimodal configuration (XGBoost-Gradient Boosting-Extra Trees) achieved superior performance across all metrics (cognitive load: $M = 68.5\%$, $SD = 2.0\%$ short, $M = 64.0\%$, $SD = 1.8\%$ long; valence: $M = 65.5\%$, $SD = 1.9\%$ short, $M = 61.5\%$, $SD = 1.7\%$ long; arousal: $M = 64.5\%$, $SD = 1.8\%$ short, $M = 60.5\%$, $SD = 1.6\%$ long). Alternative trimodal combinations maintained robust performance, with the second-best configuration (Gradient Boosting-XGBoost-Extra Trees) showing only marginally lower results (within 1.0% difference), suggesting the stability of multimodal integration.

Session Length Impact Analysis: Systematic analysis of session duration effects revealed consistent patterns. Short sessions (30 seconds) demonstrated superior performance across all configurations, with an average improvement of 4.5% ($SD = 0.8\%$) over long sessions. This advantage was most pronounced in cognitive load recognition (mean difference = 4.7%, $SD = 0.9\%$) and remained consistent across affect recognition tasks (valence: mean difference = 4.3%, $SD = 0.8\%$; arousal: mean difference = 4.2%, $SD = 0.7\%$).

Comprehensive Statistical Validation: Statistical analysis through paired t-tests confirmed the significance of our findings [57]: (1) Multimodal vs. Unimodal: All comparisons showed significant improvements ($t(38) = 4.82, p < 0.001$,

TABLE VI
ML MODEL PERFORMANCE ON COAFFINITY DATASET

Audio	Visual	Physiological	Cognitive Load		Valence		Arousal	
			Short	Long	Short	Long	Short	Long
Unimodal Models								
<i>Audio Only</i>								
XGBoost	-	-	61.5	57.0	58.0	54.5	57.5	53.0
Gradient Boosting	-	-	62.0	57.5	58.5	55.0	58.0	53.5
Extra Trees	-	-	59.0	54.5	56.5	53.0	56.0	51.5
<i>Visual Only</i>								
-	XGBoost	-	62.5	58.0	59.5	56.0	58.5	54.0
-	Gradient Boosting	-	63.0	59.5	60.5	57.0	59.5	56.0
-	Extra Trees	-	60.5	56.0	57.5	54.0	56.5	52.5
<i>Physiological Only</i>								
-	-	XGBoost	60.0	55.5	57.0	53.5	56.5	52.0
-	-	Gradient Boosting	59.5	55.0	56.5	53.0	56.0	51.5
-	-	Extra Trees	58.5	54.0	56.0	52.5	55.5	51.0
Multimodal Models								
XGBoost	Gradient Boosting	Extra Trees	68.5	64.0	65.5	61.5	64.5	60.5
Gradient Boosting	XGBoost	Extra Trees	67.5	63.0	64.5	60.5	63.5	59.5
Extra Trees	XGBoost	Gradient Boosting	67.0	62.5	64.0	60.0	63.0	59.0
XGBoost	Extra Trees	Gradient Boosting	68.0	63.5	65.0	61.0	64.0	60.0

The best-weighted F1 scores in each subcategory (audio-only, visual-only, physiological-only, and audio-visual-physiological) are underlined, and the best scores of each label are highlighted in bold.

TABLE VII
DEEP LEARNING MODEL PERFORMANCE ON COAFFINITY

Audio	Visual	Physiological	Params	Cognitive Load		Valence		Arousal	
				Short	Long	Short	Long	Short	Long
Random Classifier			-	51.6	42.0	49.3	41.2	53.42	37.0
Unimodal Models									
<i>Audio Only</i>									
ResNet-18 [61]	-	-	11.7M	62.0	55.4	61.5	58.0	59.0	51.5
wav2vec-Trans. [62]	-	-	23.0M	61.5	53.5	63.0	59.5	64.5	60.0
1D CNN-Attn. [63]	-	-	8.0M	68.5	64.0	65.5	61.5	61.5	52.5
<i>Visual Only</i>									
-	ResNet3D [64]	-	33.4M	66.0	62.5	64.5	61.0	63.5	60.0
-	ViT-TCN [65]	-	86.0M	67.0	63.5	65.0	61.5	64.0	61.0
-	Spatio-Temp. [66]	-	21.0M	68.5	64.5	66.0	62.0	65.5	61.5
<i>Physiological Only</i>									
-	-	1D ResNet [61]	10.0M	71.0	60.5	72.5	58.0	69.0	58.0
-	-	1-D Enc. CNN. [47]	7.0M	66.5	62.5	64.0	60.0	64.5	60.5
-	-	Physio Trans. [67]	90.0M	56.5	42.5	54.0	40.0	54.5	40.5
Multimodal Models									
1D CNN-Attn	Spatio-Temp.	1D ResNet	80.0M	75.0	69.5	73.0	67.5	70.5	65.0
1D CNN-Attn	Spatio-Temp.	1-D Att CNNs.	76.0M	82.6	71.4	80.2	69.5	72.1	63.5
Trans. Fusion	Trans. Fusion	Trans. Fusion	146.0M	65.0	49.5	63.0	47.5	60.5	42.0

Best-weighted F1 scores per subcategory are underlined, and the highest per label is bolded. Here, the random classifier is a randomly initialized parameter model, serving as a baseline to contextualize the performance of trained models.

$Cohen'sd = 0.78$) (2) Short vs. Long sessions: Consistent significant differences ($t(38) = 3.94, p < 0.001, Cohen'sd = 0.63$)

These comprehensive results validate the effectiveness of our multimodal approach in remote collaborative settings, particularly for short-duration analysis windows. The systematic improvement patterns across modality combinations and session lengths provide strong evidence for the robustness of our methodology.

B. Deep Learning Models

Training: The deep learning experiments implemented a comprehensive evaluation framework across unimodal, bimodal, and multimodal architectures. Models were developed using PyTorch [58], incorporating an AdamW optimizer [59] with an initial learning rate of $1e-4$ and weight decay of 0.01. Training proceeded over 20 epochs with a 5-epoch warmup period, utilizing a cosine annealing learning rate schedule to optimize convergence. To address class imbalance inherent in our dataset, we employed weighted categorical cross-entropy

loss, with early stopping (patience=5) monitoring validation loss to prevent overfitting.

Data preprocessing was carefully tailored to each modality's characteristics. Audio inputs were transformed into mel-spectrograms using 128 mel bands, employing a 25 ms window with 10 ms hop length to capture temporal dynamics effectively. Visual data underwent face detection and alignment using FaceNet [60], followed by scaling to 112×112 pixels to maintain consistent spatial dimensions. Physiological signals (PPG and GSR) were processed using a sliding window approach with 256 samples and 50% overlap, incorporating signal normalization to standardize the input distribution.

The training infrastructure used NVIDIA RTX A100 GPUs, employing single GPU configurations for unimodal models and dual GPU setups for multimodal architectures. Batch sizes were optimized to 32 for unimodal training and 16 for multimodal approaches, with gradient accumulation over 4 steps to maintain effective batch statistics.

Model Architecture Selection: Our architectural choices were guided by a literature review and empirical validation in

related domains. For audio processing, we implemented three distinct architectures: ResNet-18 [61] serving as a robust baseline, a wav2vec-transformer [62] using pre-trained representations for enhanced feature extraction, and a specialized CNN-Attention network [63] designed to capture temporal dependencies in emotional speech patterns.

Visual processing employed a complementary set of architectures: ResNet3D [64] for spatio-temporal feature learning, Vision Transformer-TCN [65] for attention-based processing, and a dedicated spatio-temporal network [66] optimized for emotion recognition in video sequences. The physiological signal processing pipeline used a 1D ResNet architecture, complemented by a 1D autoencoder-CNN framework [47] and a Physiological Transformer [67] specifically designed for biosignal analysis.

Multimodal integration employed a late fusion strategy, where modality-specific embeddings were combined through a cross-attention mechanism before processing by an MLP head. The fusion network comprised two fully connected layers with ReLU activation [68] and dropout regularization [69] ($p = 0.5$), optimizing the integration of complementary features from different modalities.

Evaluation Protocol: The evaluation framework employed a Leave-Four-Pair-Out (LFPO) cross-validation strategy to ensure a robust assessment of model generalization similar to the machine learning evaluation protocol in the section V-A. This approach systematically held out four participant pairs in each iteration for testing while training the remaining pairs, which is crucial for maintaining participant independence between training and testing sets. Performance was evaluated using weighted F1-scores to account for class imbalance in the dataset. We analysed both short (30-second) and long (2–10 minute) sessions to understand the impact of temporal window size on recognition performance. Statistical validation employed paired t-tests with Bonferroni correction for multiple comparisons, complemented by effect size calculations (Cohen's d) [57]. The training was conducted on NVIDIA RTX A100 GPUs, utilizing single-GPU configurations for unimodal models and dual-GPU setups for multimodal architectures.

1) Deep Learning Results: In unimodal configurations, each modality demonstrated distinct strengths and patterns. The Audio Attention network achieved superior performance for cognitive load recognition (68.5% F1-score in short sessions, 64.0% in long sessions), significantly outperforming the ResNet-18 baseline ($p < 0.001$). Visual processing through the Spatio-Temporal network showed robust performance (68.5% F1-score for cognitive load), particularly effective in capturing facial expressions and temporal dynamics. The 1D ResNet processing physiological signals demonstrated the strongest individual modality performance (71.0% F1-score for cognitive load in short sessions), highlighting the importance of physiological indicators for affective state recognition.

The optimal multimodal configuration, combining Audio Attention, Spatio-Temporal, and 1D Attention CNN networks, achieved the highest overall performance: 82.6% F1-score for cognitive load, 80.2% for valence, and 72.1% for arousal in short sessions. This architecture (76.0 M parameters) demonstrated superior performance while maintaining reasonable

computational efficiency compared to the full transformer-based approach (146.0 M parameters). Statistical analysis through bootstrapped confidence intervals (1000 iterations) confirmed the significance of these improvements ($p < 0.001$).

Modality Importance Analysis: We conducted systematic ablation studies to quantify each modality's contribution to the overall performance of our best-performing model (Audio Attention + Spatio-Temporal + 1-D Att CNNs, 82.6% F1-score). By individually removing each modality and measuring performance degradation, we found physiological signals contributed most significantly (31.2% contribution to F1-score), followed by visual (28.7%) and audio (25.4%) modalities. The remaining 14.7% represents the synergistic effects of multimodal fusion.

Statistical validation through paired t-tests confirmed the significance of these contributions ($t(38) = 12.45$, $p < 0.001$, Cohen's $d = 0.82$). Cross-modal correlation analysis revealed the strongest feature alignment between audio and visual modalities ($\rho = 0.68 \pm 0.05$), while physiological signals showed moderate correlations with both audio ($\rho = 0.45 \pm 0.07$) and visual ($\rho = 0.42 \pm 0.06$) features. This aligns with our results where the Audio Att. + Spatio-Temporal + 1-D Att. CNN's architecture achieved superior performance by effectively leveraging these complementary modality relationships.

VI. DISCUSSION

a) Model Performance: Our analysis of traditional ML and DL approaches for cognitive load and affect recognition in remote collaborative settings reveals several significant insights. The integration of multimodal data streams, particularly the novel combination of physiological signals with audio-visual information, demonstrates substantial performance improvements over single-modality approaches. The DL models consistently outperformed traditional machine learning approaches, with the best-performing DL configuration (Audio Attention + Spatio-Temporal + 1D Attention CNN) achieving an 82.6% F1-score for cognitive load recognition compared to 68.5% for the best ML model (XGBoost-Gradient Boosting-Extra Trees). This performance differential can be attributed to DL's superior ability to automatically learn hierarchical representations and capture complex temporal dynamics through attention mechanisms.

Analysis of modality importance revealed physiological signals as crucial contributors across both paradigms, with DL showing a 31.2% contribution to overall performance and ML demonstrating significant improvement in recognition accuracy (4–6%). The effectiveness of physiological signals was particularly pronounced in short-term cognitive load assessment, suggesting their importance in future affective computing systems. This finding is especially relevant given the increasing prevalence of wearable sensors in everyday settings. The audio-visual modalities showed complementary strengths, with visual features achieving highest individual accuracy in ML approaches and audio attention mechanisms demonstrating superior temporal modelling in DL methods.

A consistent finding across both ML and DL approaches was the superior performance in short-term analysis windows, with ML showing an average improvement of $4\% \pm 0.8\%$

and DL showing $3.8\% \pm 0.7\%$ for short versus long sessions. This consistency across computational approaches suggests a fundamental characteristic of affective state recognition rather than a limitation of specific computational methods. The temporal analysis revealed that short-term windows (30 seconds) effectively capture immediate emotional and cognitive states and improve continuous prediction and monitoring applications.

b) Group Synchronisation: Group Synchronisation analysis of both PPG and EDA revealed clear task-dependent patterns of group synchrony: socially oriented activities (*Open Discussion*, *Lighten the Mood*) displayed significantly higher coherence, whereas cognitively or spatially intense tasks (*Map Matching*, *Multitask*) showed marked decreases. These findings align with prior research demonstrating that group arousal converges more readily in supportive or humour-driven scenarios, whereas individual attentional demands and stress responses dominate under high-load conditions [20], [51]. Specifically, increased PPG and EDA coherence suggests that shared emotional engagement fosters stronger interpersonal alignment, while puzzle-like or multitasking settings cause physiological divergence as each member concentrates on distinct goals. Consequently, coherence metrics can serve as reliable indicators of whether group members are collectively experiencing positive or stressful states, providing practical implications for designing interventions to promote cohesive remote teamwork [19]. By tracking these synchronization indices in real-time, group-based interfaces could adapt task difficulty, introduce pauses, or offer emotional support strategies, thereby optimizing both performance and well-being [70].

The practical implications of our findings extend to several domains. For remote collaboration tools, our results suggest the integration of multimodal sensing for a better understanding of participant states, with particular emphasis on physiological signal integration [3], [37]. The superior performance of both ML and DL models in short-term analysis windows suggests implementing sliding window approaches for real-time [17]. Adaptive interfaces could benefit from this research by incorporating real-time cognitive load detection, potentially adjusting meeting dynamics or user interfaces based on participant states [37]. In future work, we will explore subject-dependent evaluation, as it could offer valuable insights for personalized models, balancing the trade-off between generalization and individual adaptability.

VII. LIMITATIONS AND FUTURE WORK

While CoAffinity offers a comprehensive and multimodal dataset for cognitive load and affective state analysis, several limitations must be addressed. First, its 39-participant sample may limit generalization. Second, our controlled tasks might not fully mirror real-world remote work; more naturalistic or longitudinal approaches could improve ecological validity. Third, exploring advanced architectures (e.g., transformers), various pre-training methods, and cross-dataset comparisons (e.g., with AVCAffe) could further validate the findings. Fourth, self-reported data are subjective; pairing them with objective metrics and external ratings would offer a fuller perspective. Fifth, group synchrony was not fully integrated—future studies

should consider such contextual factors. Lastly, the focus on specific affective states and cognitive load metrics could be broadened to capture a wider emotional and cognitive range.

CoAffinity can guide adaptive systems that gauge users' stress or engagement in real-time, potentially improving remote collaboration and mental health support [70]. Researchers can combine CoAffinity with complementary datasets (e.g., AVCAffe) for robust cross-domain validation and performance benchmarking [51]. Such efforts may accelerate progress in affective computing, human–AI teaming, and remote work solutions, supporting interventions that address cognitive overload, burnout, and team cohesion in both academic and industry applications [2]. Ultimately, extending CoAffinity's demographic coverage and methodological breadth will deepen our theoretical understanding of group cognition and emotion, while enabling advanced design of empathic remote work environments [1].

VIII. CONCLUSION

CoAffinity is a novel multimodal corpus that unites audio, video, and physiological data (PPG and GSR) from 39 participants engaged in simulated remote work tasks. By annotating cognitive load and affect, this dataset advances affective computing research on how multimodal signals can improve the detection of cognitive and emotional states. Our findings show that combining multiple modalities significantly enhances predictive accuracy while also highlighting the role of physiological synchrony in shaping collaborative outcomes. Group synchrony in PPG and GSR can influence both emotional experiences and cognitive performance, indicating clear opportunities for adaptive, empathy-driven collaboration tools. Although limited by its relatively small sample and a controlled laboratory setting, CoAffinity enhances our understanding of cognitive load, emotional states and group synchronisation in the design of human-centred remote collaboration.

DATASET AND CODE AVAILABILITY

The dataset is available to researchers after the signing of a sharing agreement. Please contact the first author for more information.

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